

CUNNAMULLA, AustraliaWMO: **945000**Lat: **28.071S**Lon: **145.681E**Elev: **625**StdP: **14.37**Time Zone: **10.00 (E10)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.1	41.2	19.3	14.8	70.5	22.6	17.4	67.7	17.8	68.6	14.5	63.0	2.3	180	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.7	103.9	69.3	101.5	68.8	99.1	68.3	76.5	91.1	74.9	88.5	73.5	86.7	6.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	126.4	80.8	71.5	118.9	80.2	69.5	111.1	79.3	40.5	91.1	38.9	88.3	37.5	86.0	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.5	11.8	DB	34.6	108.7	2.1	2.8	33.1	110.7	31.9	112.3	30.8	113.9	29.2	115.9	(4)
(5)			WB	32.4	80.3	2.4	4.5	30.7	83.6	29.4	86.2	28.0	88.7	26.3	92.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	70.8	85.4	83.1	79.1	70.5	62.3	56.7	55.4	58.9	66.4	72.0	77.6	83.0	(6)
(7)		DBStd	11.99	5.90	5.96	5.38	5.88	5.18	5.67	4.88	6.16	7.17	6.14	7.12	5.78	(7)
(8)		HDD50	15	0	0	0	0	0	5	7	3	0	0	0	0	(8)
(9)		HDD65	980	0	0	0	18	114	257	300	211	68	11	1	0	(9)
(10)		CDD50	7607	1096	927	902	615	382	206	174	278	493	683	829	1022	(10)
(11)		CDD65	3097	631	507	437	183	31	8	2	22	110	229	380	557	(11)
(12)		CDH74	39037	8885	6717	5036	1761	193	47	11	254	1315	2621	4598	7599	(12)
(13)		CDH80	20743	5369	3784	2469	632	16	0	0	56	478	1164	2384	4391	(13)
(14)	Wind	WSAvg	4.5	4.9	5.1	4.4	3.8	3.7	3.9	3.9	4.3	4.7	4.9	5.5	5.1	(14)
(15)	Precipitation	PrecAvg	15.1	2.5	1.6	1.6	0.9	1.4	0.8	0.9	0.8	0.6	1.0	1.4	1.4	(15)
(16)		PrecMax	28.4	9.6	9.7	6.7	4.1	5.9	3.4	4.4	3.4	2.6	4.3	5.4	5.5	(16)
(17)		PrecMin	4.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	5.4	2.4	1.8	1.7	1.2	1.5	0.8	1.1	0.7	0.7	1.0	1.4	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	108.2	105.6	101.0	92.7	82.5	78.6	75.4	86.7	94.3	98.3	104.6	104.2	(19)
MCWB			69.8	71.6	68.8	67.6	61.8	61.7	57.1	62.2	62.0	63.8	64.7	68.0	(20)	
2%		DB	104.8	102.0	96.7	89.5	79.3	75.6	72.1	80.3	89.5	94.1	99.9	102.0	(21)	
		MCWB	70.3	70.7	68.3	64.2	61.2	60.3	55.5	58.8	61.1	63.4	64.8	68.0	(22)	
5%		DB	102.0	99.1	93.9	86.7	76.4	71.9	69.7	76.3	86.2	90.8	95.8	99.7	(23)	
		MCWB	69.9	70.5	68.1	63.3	60.1	57.8	54.4	56.8	60.4	63.4	65.2	67.6	(24)	
10%		DB	99.0	95.9	91.2	83.6	73.4	68.1	66.4	72.3	82.2	87.1	92.1	97.2	(25)	
		MCWB	69.1	69.9	67.1	62.0	59.1	56.2	52.8	55.3	58.7	61.7	65.0	66.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.1	78.9	76.1	73.3	68.7	64.1	62.1	64.5	67.3	71.9	74.5	76.1	(27)
MCD B			95.8	91.1	86.1	86.2	71.1	74.5	68.2	81.7	77.5	87.6	86.3	93.8	(28)	
2%		WB	76.7	76.3	73.9	69.5	65.6	61.8	58.5	61.9	64.2	68.2	71.9	73.6	(29)	
		MCD B	93.2	89.9	84.4	78.2	73.6	71.1	68.4	75.7	79.8	81.6	83.5	89.4	(30)	
5%		WB	74.7	74.9	71.9	66.7	63.5	60.0	56.6	59.3	62.7	66.0	70.3	71.7	(31)	
		MCD B	89.3	88.6	85.0	76.9	72.1	69.1	66.1	72.0	80.5	80.3	81.9	87.7	(32)	
10%		WB	73.0	73.5	70.2	64.6	61.6	57.9	54.4	57.0	61.0	64.2	69.0	70.1	(33)	
		MCD B	87.7	85.9	84.5	77.8	70.0	66.3	63.6	69.0	78.0	79.8	81.0	89.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.7	20.7	21.8	22.2	19.8	20.6	22.0	24.1	25.3	24.0	21.8	22.7	(35)
MCDBR			25.3	24.8	24.5	25.8	23.1	23.4	25.2	27.8	28.3	29.4	25.7	25.6	(36)	
MCWBR			8.0	7.9	8.9	10.2	10.5	11.3	12.3	12.7	11.1	11.3	9.3	8.7	(37)	
5% WB		MCD BR	20.7	20.0	18.7	19.2	17.0	20.2	20.1	23.5	24.7	22.2	19.1	21.0	(38)	
		MCWBR	7.9	8.5	7.9	9.2	8.5	10.8	11.3	11.6	10.7	11.3	8.2	8.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.355	0.350	0.331	0.315	0.297	0.288	0.286	0.299	0.338	0.344	0.358	0.354	(40)	
taud		2.520	2.550	2.595	2.589	2.593	2.621	2.603	2.541	2.425	2.436	2.450	2.490	(41)		
Ebn at Noon		313	309	306	296	287	284	289	297	298	307	310	314	(42)		
Edh at Noon		36	34	30	28	25	23	24	29	36	37	38	37	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2419	2237	1968	1626	1274	1059	1197	1515	1880	2204	2328	2439	(44)	
(45)		RadStd	121	155	111	93	58	70	75	74	115	143	204	162	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (2 neighbors)	+1.19	N/A	N/A	+1.58	N/A	N/A	N/A	N/A	N/A	+299	(47)

Nomenclature: See separate page